Sample output:-

packet sniffer started...
waiting for packets...
- - - - - - - - - -
Source Ip : 192.168.1.10
Destination Ip : 142.250.183.14
protocol : 6
packet size : 60 bytes
- - - - - - - - - -
Source Ip : 8.8.8.8
Destination Ip : 192.168.1.10
Protocol : 1
Packet Size : 84 bytes
- - - - - - - - - -

Result:-

Thus, the ping program and packet sniffing program were successfully implemented using python raw sockets.

Exp. No. 14

Packet Sniffing using Raw Socket.

Aim:-

To implement a packet sniffing using python raw socket and display the source Ip, destination Ip and protocol of captured packets.

Code:-

```
import socket
import struct
sock = socket.socket(
        socket.AF_INET
        socket.sock_RAW,
        socket.IPPROTO_IP)
print("packet sniffer started...")
print("Waiting for packets...\n")
while True:-
    packet, address = sock.recvfrom(65535)
    ip_header = packet[0:20]
    ip = struct.unpack(
            "!BBHHHBBH4s4s",
            ip_header)
    source_ip = socket.inet_ntoa(iph[8])
    destination_ip = socket.inet_ntoa(iph[9])
    protocol = iph[6]
    print("- - - - - - - - -")
    print("Source Ip:", source_ip)
    print("Destination Ip:", destination_ip)
    print("Protocol:", protocol)
    print("Packet size:", len(packet), "bytes")
    print("- - - - - - - - -")
```